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Comparative Study of Temporal Segmentation Methods for Radar-Based Kick Gesture Recognition

Project Thesis

by

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Abstract

Radar-based gesture recognition has gained increasing attention as a robust and privacy preserving alternative to camera-based human - machine interaction systems, particularly in automotive and smart environment applications. Kick-based foot gestures are particularly useful for situations involving hands-free engagement, such opening a trunk or controlling access, among other dynamic gestures. However, considerable inter-person variability, background motion, and the requirement for dependable temporal segmentation of radar signals continue to make accurate recognition of kick motions difficult.

This thesis uses frequency-modulated continuous-wave (FMCW) radar data to examine and contrast several signal segmentation techniques for kick gesture recognition. In order to stabilize signal amplitudes across subjects and recording sessions, a full processing pipeline is created, beginning with raw in-phase and quadrature (I/Q) radar signals. This is followed by band-pass filtering and global normalization. Three segmentation techniques energy-based peak segmentation, gradient-based peak segmentation, and Doppler-based segmentation using short-time Fourier transform (STFT) analysis are investigated in order to extract useful temporal segments from lengthy radar recordings.

A one-dimensional convolutional neural network (CNN) is trained for binary classification between legitimate kick gestures and non-kick motions using fixed-length data windows of 4000, 2000, and 1000 samples for each segmentation technique. A person-wise data split is used, with a predetermined selection of test participants omitted from training, to guarantee subject-independent evaluation. Accuracy, precision, recall, false positive rate, area under the receiver operating characteristic curve (AUC), normalized confusion matrices, and training convergence curves are used to assess the model's performance.

The results of the experiments show that window length and segmentation approach significantly affect classification performance. While gradient-based and Doppler-based techniques demonstrate better temporal localization for shorter windows, energy-based segmentation offers consistent performance for larger windows. In particular, the Doppler-based method uses frequency-domain analysis to successfully capture motion dynamics, which improves the ability to distinguish between kick and non-kick movements in difficult situations.

Overall, this work emphasizes the significance of integrating suitable temporal segmentation with deep learning models and offers a methodical comparison of segmentation strategies for radar-based kick gesture recognition. The results provide useful information for the development of dependable radar-based gesture interfaces in actual settings.

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Chapter 1 - Introduction

1.1 Background and Motivation

In contemporary human-machine interfaces, contactless interaction systems have grown in significance, especially in automotive and smart access applications where dependability, comfort, and hygiene are crucial. One well-known example is kick-to-open vehicle access, in which a user kicks or places their foot beneath a car to initiate an automated action, like opening a door or trunk. When the user's hands are busy, these kick movements offer an easy-to-use, hands-free engagement option.

When it comes to lower-limb gesture identification, traditional gesture detection sensor technologies like cameras, infrared sensors, and pressure-based switches have a number of drawbacks. Camera sensors can provide privacy issues, as vision-based systems are susceptible to occlusions, shadows, and lighting. Because of temperature fluctuations, surface reflections, and restricted sensing range, infrared and ultrasonic sensors frequently have trouble in outdoor settings. For strong kick gesture identification in real-world vehicle applications, these difficulties render such sensors unreliable.

Because radar-based sensing is lighting-invariant, resilient to environmental changes, and privacy-preserving, it has become a potent substitute for motion and gesture detection. Specifically, the in-phase (I) and quadrature (Q) signal components of FMCW (Frequency-Modulated Continuous-Wave) radar allow for the acquisition of fine-grained motion information. Radar is particularly useful for identifying dynamic lower-limb motions like kicks because these signals capture both amplitude and phase fluctuations brought on by moving objects.

1.2 Problem Statement

Despite the fact that radar sensors are ideal for detecting kick gestures, accurate recognition is still difficult for a number of reasons:

- Kick execution techniques, speeds, and motion trajectories vary greatly across individuals.
- Temporal variability is the variation in kick gesture duration and start between repeats.
- Non-gesture actions, such strolling by the car or standing, might cause radar reflections that resemble kick gestures.
- continuous radar recordings in which only a brief portion of a longer signal window is occupied by the pertinent kick motion.

Therefore, a crucial step in accurately recognizing kick gestures is the efficient temporal segmentation of radar signals. Even with strong deep learning models, poor segmentation might introduce irrelevant signal sections, dilute discriminative features, and deteriorate classification performance.

1.3 Research Objectives

This thesis main goal is to develop and assess a radar-based kick gesture detection framework that blends deep learning-based classification with signal-level segmentation methods.

This effort specifically seeks to:

- Create and contrast three segmentation techniques that separate continuous radar signals from kick gestures:
  - **Energy Peak Segmentation**
  - **Gradient Peak Segmentation**
  - **Doppler (STFT) Peak Segmentation**
- Examine how recognition ability is affected by varying temporal window lengths (4000, 2000, and 1000 samples).
- Create a Convolutional Neural Network (CNN) that can use segmented I/Q radar signals to learn discriminative spatiotemporal characteristics.
- To evaluate generalization across unseen users, employ fixed person-wise train-test splits in a person-independent manner.
- Using performance indicators like accuracy, precision, recall, false positive rate, and ROC-AUC, present a thorough comparison.

1.4 Contributions of This Thesis

The key contributions of this thesis can be summarized as follows:

1. **A complete radar signal processing pipeline** for kick gesture recognition, including filtering, normalization, segmentation, and classification.
2. **A comparative study of three physically motivated segmentation techniques**, each capturing different characteristics of kick motion dynamics:
   - Energy-based motion intensity
   - Gradient-based motion onset
   - Doppler-based velocity distribution
3. **A systematic analysis of window length selection**, highlighting the trade-off between temporal resolution and classification performance.
4. **Person-independent experimental evaluation**, demonstrating the robustness of the proposed methods for real-world deployment.
5. **Thesis-style visualizations**, including signal plots, segmentation overlays, training curves, ROC curves, and confusion matrices, to support transparent and interpretable analysis.

Chapter 2 - Radar-Based Gesture Recognition – Related Work

2.1 Overview of Radar-Based Gesture Recognition

In recent years, radar-based sensing has drawn a lot of attention for gesture and motion identification applications because of its resilience to environmental changes and ability to protect privacy. Radar sensors are especially useful for interior and automobile applications since, in contrast to vision-based systems, they function irrespective of lighting conditions and do not record recognizable visual information.

Hand and arm motions were the main focus of early radar gesture recognition research, which took advantage of micro-Doppler signals produced by moving limbs. These research showed that rich motion information found in radar signals can be used for categorization using both learning-based methods and manually created features. However, because lower-limb motions had distinct velocity profiles, amplitudes, and spatial properties, many of these systems were not directly applicable to kick or foot gestures.

Radar sensors are usually installed near the ground in car access systems like kick-to-open trunks, and the lower body is where the majority of the reflections come from. Additional difficulties brought forth by this arrangement include debris on the ground, interference from walking motions, and inconsistent kick execution. Therefore, for accurate kick gesture recognition, specific processing and segmentation techniques are needed.

2.2 Kick and Lower-Limb Gesture Recognition

Kick and foot-based gesture identification has gotten comparatively less attention in the literature than hand gesture recognition. However, a number of studies have looked into the use of radar and other sensing modalities for lower-limb motion detection.

For kick detection, some research has looked into pressure sensors or capacitive proximity sensors installed in car bumpers. Although these methods are capable of identifying the presence of a foot, they frequently lack resilience to false activations brought on by unintentional movements or environmental disruptions. Furthermore, it can be challenging to differentiate purposeful kicks from other lower-body motions because these sensors usually provide limited temporal information.

By recording continuous time-series data that captures the dynamic motion pattern of a kick, encompassing approach, impact, and retraction phases, radar-based methods get beyond these restrictions. Kick movements produce unique energy bursts and Doppler patterns that are different from walking or standing motions, according to research. These results encourage the application of radar signal processing methods that specifically take advantage of temporal and frequency-domain features.

2.3 Signal Preprocessing for Radar Gesture Recognition

2.3.1 Bandpass Filtering

Bandpass filters are used in a number of studies to minimize high-frequency noise and low-frequency congestion while isolating the frequency range linked to human motion. The radar operating frequency and sampling rate are taken into consideration while choosing typical cutoff frequencies. Bandpass filtering is used in kick gesture identification to retain motion-induced elements from the moving foot while suppressing static reflections from the ground and vehicle body.

2.3.2 Normalization

To lessen variability between participants and recording sessions, amplitude normalization is frequently used. In deep learning pipelines, global min–max normalization to a fixed range, like [−1,1] is frequently employed to stabilize training and stop some signals from dominating because of higher amplitudes. In radar-based kick gesture datasets, where reflection strength can vary greatly according on footwear, distance, and execution style, normalization is especially crucial.

2.4 Temporal Segmentation of Radar Signals

Temporal segmentation is one of the most important issues in radar-based kick gesture identification. While the actual kick motion only takes up a brief temporal interval, radar recordings can contain lengthy continuous signals. Precise segmentation guarantees that the classifier avoids dilution by irrelevant input and concentrates on useful signal regions.

2.4.1 Fixed Sliding Window Segmentation

In order to extract segments from continuous radar data, early methods frequently used fixed sliding windows with overlap. Despite being easy to use, this approach frequently generates a lot of duplicated or uninformative segments, which raises computational costs and may result in label noise. Furthermore, sliding windows do not adjust to changes in the timing or intensity of gestures.

2.4.2 Energy-Based Segmentation

By calculating short-time energy or signal magnitude, energy-based segmentation algorithms pinpoint areas of interest. These techniques make the assumption that meaningful gestures match signal energy peaks brought on by quick limb movement. Energy-based methods have been effectively used in radar gesture identification and audio processing because they are computationally efficient. This approach is especially useful for kick gestures since energy peaks frequently coincide with the moment of foot acceleration or impact.

2.4.3 Gradient-Based Segmentation

Instead of focusing on absolute energy, gradient-based segmentation looks for quick changes in signal magnitude. These techniques seek to capture the beginning of motion by examining the temporal derivative of the signal, which is particularly important for kick gestures with a distinct commencement period. When absolute energy differs between subjects, gradient-based algorithms can perform better than energy-based ones because they are more sensitive to motion transitions.

2.4.4 Doppler-Based Segmentation

Doppler-based segmentation makes use of radar signals' frequency-domain form. Researchers extract Doppler spectrograms, which show velocity patterns across time, using methods like the Short-Time Fourier Transform (STFT). Kick motions generate unique Doppler energy patterns because the foot moves quickly forward and backward. These techniques offer a physically significant segmentation mechanism that is resilient to amplitude fluctuations by locating peaks in Doppler energy.

2.5 Feature Learning and Classification

2.5.1 Handcrafted Features

Previous radar gesture detection systems mostly depended on manually created features from micro-Doppler descriptors, frequency-domain energy distributions, or time-domain data. Although these qualities work well in controlled situations, they frequently need to be carefully adjusted and may not translate well to different subjects or environments.

2.5.2 Deep Learning Approaches

Data-driven feature learning has become more popular as a result of recent developments in deep learning. Because Convolutional Neural Networks (CNNs) can learn hierarchical representations from raw or slightly processed inputs, they have been frequently used for radar gesture identification.

CNNs that process time-series I/Q data are able to identify cross-channel correlations, phase relationships, and local temporal patterns. CNNs are especially well suited for kick gesture identification because they don't need explicit feature engineering to simulate the intricate temporal structure of lower-limb motion.

2.6 Evaluation Strategies and Person Independence

Due to subject overlap between training and testing sets, many current research assess gesture recognition ability using random train–test splits, which may produce optimistic results. Person-independent evaluation, on the other hand, offers a more accurate assessment of real-world performance since test subjects are totally hidden during training.

The significance of person-wise splitting for gesture recognition systems meant for deployment is highlighted in a number of recent studies. Since individual variations in leg length, movement speed, and style can greatly affect radar signatures, this evaluation method is particularly pertinent to kick gestures.

2.7 Summary and Research Gap

From the reviewed literature, the following observations can be made:
Radar sensing is well suited for kick and foot gesture recognition, particularly in automotive contexts.

- Temporal segmentation remains a key challenge, with energy-, gradient-, and Doppler-based methods offering complementary strengths.
- Deep learning models, especially CNNs, outperform handcrafted feature-based approaches when sufficient data and proper segmentation are employed.
- Person-independent evaluation is essential for assessing generalization but is not consistently adopted in existing studies.

Research Gap: Despite these developments, little research has been done that systematically analyzes various segmentation algorithms for kick gesture detection using fixed person-wise splits, multiple window widths, and a consistent CNN architecture under equivalent experimental settings. By offering a cohesive comparative approach based on both deep learning and signal processing principles, this thesis fills this gap.

Chapter 3 - Data Acquisition and Preprocessing

3.1 Overview of the Experimental Setup

The radar dataset utilized in this thesis and the entire preprocessing workflow used prior to segmentation and classification are described in this chapter. Preprocessing aims to transform raw radar readings into a clean, standardized, and organized representation that can be used for kick gesture recognition using deep learning.

Radar I/Q time-series signals from several participants executing both kick-based and non-kick motions make up the dataset. In order to replicate a genuine car kick-to-open scenario, each recording records reflections from the lower body while the radar sensor is placed to watch foot action near the ground.

3.2 Radar Signal Acquisition

3.2.1 Radar Modality and Signal Representation

The radar system provides complex baseband signals, represented by In-phase (I) and Quadrature (Q) components. For each recording, four synchronized channels are available:

- $I_{lower}(t)$
- $Q_{lower}(t)$
- $I_{upper}(t)$
- $I_{upper}(t)$

These channels correspond to different antenna paths or range bins within the radar system and collectively capture the spatio-temporal dynamics of the kick gesture.

Each signal is sampled at a fixed sampling frequency of:

$$\boldsymbol{f}_{\mathrm{s}} = 400\mathbf{0}\ \boldsymbol{Hz}$$

This sampling rate is sufficient to capture rapid lower-limb motion and Doppler variations caused by kicking actions.

3.3 Dataset Organization and Labeling

3.3.1 Subject-Wise Data Structure

Each participant has many recordings that match to various gesture repetitions in the person-wise organization of the collection. Realistic performance evaluation requires person-independent evaluation, which is made possible by this structure.

For each radar recording, the following metadata is assigned:

Person ID – unique identifier for The binary label is assigned as:

- each subject
- **Gesture ID** – numerical label corresponding to the performed gesture
- **Repetition index** – multiple trials per gesture
- **Combination ID** – unique identifier combining person, gesture, and repetition

The combination identifier is defined as:

$$\text{Combination} = \text{p}\langle\text{Person}\rangle - \text{g}\langle\text{Gesture}\rangle - \text{r}\langle\text{Repetition}\rangle$$

3.3.2 Binary Label Definition

For this thesis, the primary focus is on **kick gesture detection**, formulated as a **binary classification problem**:

- **Valid**: intentional kick gestures
- **Invalid**: non-kick or irrelevant lower-body motions

The binary label is assigned as:

$$y = \begin{cases} Valid, & if\ gesture \in \{0,1,2,3\} \\ Invalid, & otherwise \end{cases}$$

The system must differentiate between deliberate kicks and normal movements in a genuine automobile use case, which is reflected in this formulation.

3.4 Signal Preprocessing Pipeline

The raw radar signals contain static clutter, noise, and amplitude variations across subjects. To address these issues, a three-stage preprocessing pipeline is applied:

1. Bandpass filtering
2. Amplitude normalization
3. Structured dataset construction

Each step is detailed below.

3.5 Bandpass Filtering

3.5.1 Motivation

Among the radar signals are:

- Static reflections from the ground and vehicle body provide low-frequency components.
- Unrelated to human movements, high-frequency noise

Doppler frequencies are mostly induced within a narrow band by kick gestures. Consequently, motion-relevant components are isolated via bandpass filtering.

3.5.2 Filter Design

A **6th-order Butterworth bandpass filter** is used due to its smooth frequency response and minimal phase distortion.
The passband is defined as:

$$\boldsymbol{f_{low} = 50\ Hz,\ f_{high} = 1999\ Hz}$$

Given the sampling frequency $f_s = 4000\ Hz$, the normalized cutoff frequencies are:

$$\omega_{low} = \frac{50}{f_s/2},\ \omega_{high} = \frac{1999}{f_s/2},$$

For numerical stability, the filter is applied using zero-phase filtering and implemented in second-order sections (SOS) form:

$$x_{bp}(t) = sosfiltfilt(x(t))$$

This ensures that temporal alignment of kick events is preserved.

3.5.3 Filtering Across All Channels

The bandpass filter is applied independently to all four I/Q channels:

$$\{I_{lower}, \mathrm{Q}_{lower}, I_{\mathrm{upper}}, \mathrm{Q}_{\mathrm{upper}}\} \rightarrow \{I^{bp}_{lower}, \mathrm{Q}^{bp}_{lower}, I^{bp}_{\mathrm{upper}}, \mathrm{Q}^{bp}_{\mathrm{upper}}\}$$

This guarantees consistent preprocessing across spatial channels.

3.6 Amplitude Normalization

3.6.1 Motivation

Significant variations in radar reflection strength are caused by:

- Height and footwear of the subject
- The radar's distance
- Each person's kicking style

CNN training may be dominated by signals with higher amplitudes if normalization is not used.

3.6.2 Global Min–Max Normalization

Global min–max normalization is used for all bandpass-filtered signals to provide uniform scaling:

$$x_{norm}(t) = 2.\frac{x(t) - x_{min}}{ax - x_{min}} - 1$$

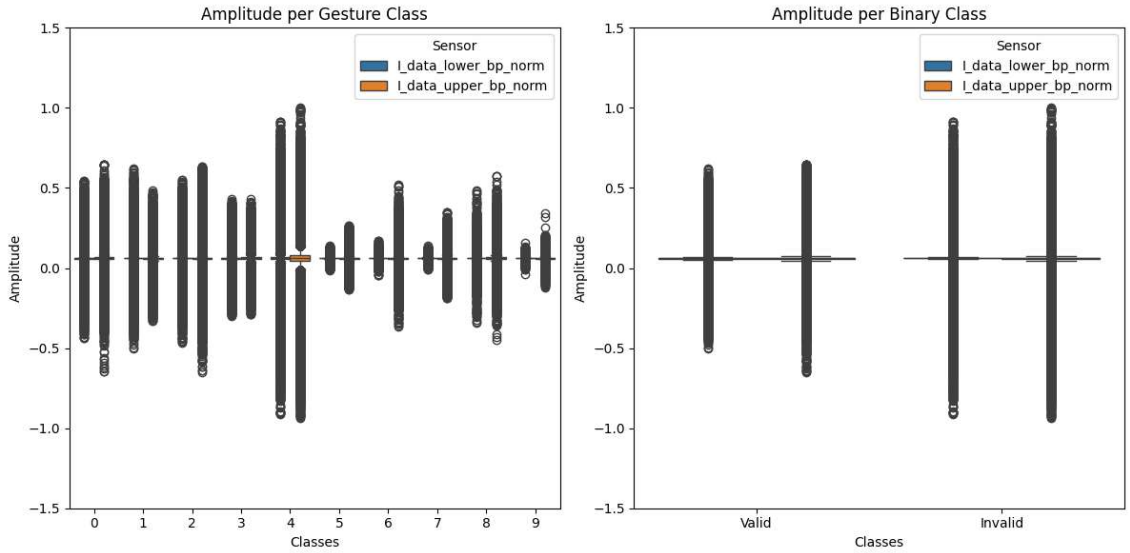
where:

$$x_{min} = min\,(X_{bp}), \qquad x_{max} = max\,(X_{bp})$$

This maps all signals into the range:

$$x_{norm}(t) \in [-1, 1]$$

To provide uniform scaling across training and testing sets, normalization is implemented once globally rather than per person.



3.7 Exploratory Signal Visualization

To verify the quality of preprocessing, exploratory visualization is carried out prior to segmentation and model training.

3.7.1 Amplitude Distribution Analysis

To examine amplitude distributions, boxplots are created:

- In all gesture classes
- Between binary labels (Invalid vs. Valid)

These plots attest to:

- Amplitudes are successfully confined by normalization.
- Because of their dynamic action, kick motions are more variable.
- Temporal modeling is necessary because the distributions of valid and invalid classes overlap.

3.8 Summary

The whole data collection and preprocessing pipeline used in this thesis was described in this chapter. Important contributions consist of:

- Person-wise construction of structured datasets
- Sturdy bandpass filtering customized for kick motion frequencies
- Stable deep learning training by global normalizing
- Initial visualization to confirm signal properties

The segmentation techniques presented in the following chapter are based on the preprocessed data.

Chapter 4 - Methodology: Kick Gesture Segmentation and Classification

4.1 Chapter Overview

The entire approach used to identify kick gestures using radar signals is presented in this chapter. The main goal is to use a deep learning model to identify each radar data by precisely locating the informative part that correlates to the kick motion.

To achieve this, three **segmentation strategies** are investigated and compared:

1. **Energy Peak Segmentation**
2. **Gradient Peak Segmentation**
3. **Doppler (STFT-based) Peak Segmentation**

To ensure a fair and controlled comparison, all three approaches use the same preprocessed radar signals and are assessed using the same CNN architecture, person-wise data partitioning, and assessment technique.

4.2 Problem Formulation

Let a radar recording be represented as a multichannel time-series:

$$\mathrm{X} = \{\mathrm{x(t)} \in \boldsymbol{R^4} \mid \mathrm{t} = 1, \ldots, \mathrm{T}\}$$

where the four channels correspond to:

$$\left[I_{lower}, \mathrm{Q}_{lower}, I_{\mathrm{upper}}, \mathrm{Q}_{\mathrm{upper}}\right]$$

The duration of each recording varies $T \approx 12000$ samples. Nevertheless, the kick motion only takes up a little amount of time within this window. As a result, the task is split into two smaller issues:

- $Temporal\ segmentation$: $Take\ a\ fixed\ length\ window\ L\ that\ is\ centered\ on\ the\ kick\ event.$
- Determine whether the segment matches to a legitimate kick gesture using binary classification

Formally, the classifier learns a mapping:

$$f_\theta : R^{L \times 4} \rightarrow \{0,1\}$$

Where $L \in \{4000,2000,1000\}$

4.3 Fixed-Length Windowing Strategy

4.3.1 Motivation

Radar recordings vary in length, while convolutional neural networks demand fixed-size inputs. Moreover:

- Noise and extraneous motion may be present in large windows.
- Small windows could overlook crucial temporal context.

Three window lengths are assessed in order to examine this trade-off.
Training and evaluation processes are kept strictly apart in this work. A single fixed-length segment ($L \in \{4000, 2000, 1000\}$) taken from each 12000-sample radar recording using the suggested segmentation method is used to train the CNN. A sliding-window technique is used for the whole 12000-sample recording during evaluation on unseen test participants, and the final prediction is derived by aggregating predictions across overlapping windows. This approach adheres to the evaluation criteria suggested in current radar-based gesture recognition research and replicates realistic deployment settings.

Window Length	Duration (approx.)	Motivation
4000 samples	~1.0 s	Full kick motion
2000 samples	~0.5 s	Compact but informative
1000 samples	~0.25 s	Minimal discriminative region

This multi-scale analysis directly follows best practices in radar-based gesture recognition literature.

4.4 Method 1: Energy Peak Segmentation

4.4.1 Rationale

The quick acceleration and deceleration of the foot during a kick gesture results in a significant increase in reflected radar energy. The goal of energy-based segmentation is to pinpoint the moment in time when this motion is at its strongest.

4.4.2 Signal Energy Computation

First, the magnitude signal is computed across all four channels:

$$m(t) = \sqrt{I_{lower}^2(t) + Q_{lower}^2(t) + I_{upper}^2(t) + Q_{upper}^2(t)}$$

Next, a **short-time energy** signal is calculated using a moving window of size F:

$$E(t) = \sum_{k = t - \frac{F}{2}}^{t + \frac{F}{2}} m^2(k)$$

In this work, $F = 50$ samples.

4.4.3 Peak Detection and Window Extraction

The kick event is assumed to occur at:

$$t^* = arg\ max\ E(t)$$

A fixed-length window is extracted as:

$$X_{seg} = X[t^* - L/2 : t^* + L/2]$$

Boundary conditions are handled to ensure valid indexing.

4.4.4 Algorithm 1: Energy Peak Segmentation

Input: Multichannel signal X, window length L
Output: Segmented signal X_seg

1: Compute magnitude m(t) across all channels
2: Compute short-time energy E(t) using moving window
3: Find peak index t* = argmax(E)
4: Extract window [t* − L/2, t* + L/2]
5: Return X_seg

4.4.5 Discussion

This approach is:

- Easy to use and computationally effective
- Strong to moderate noise
- Effective when background motion is dominated by kick energy

It might not work, though, if similar energy levels are produced by non-kick motions.

4.5 Method 2: Gradient Peak Segmentation

4.5.1 Motivation

Gradient-based segmentation, in contrast to energy-based techniques, concentrates on quick temporal changes that are typical of the foot's abrupt acceleration during a kick.

4.5.2 Gradient-Based Decision Signal

As in Method 1, the magnitude signal m (t) is first calculated. Next, the absolute temporal gradient is computed:

$$\boldsymbol{g(t)} = \left|\frac{\boldsymbol{dm(t)}}{\boldsymbol{dt}}\right|$$

To suppress noise, a smoothed gradient is obtained using a moving average filter:

$$\tilde{\boldsymbol{g}}(\boldsymbol{t}) = \frac{1}{S}\sum_{k=t-S/2}^{t+S/2} g(k)$$

where $S = 25$ samples.

4.5.3 Peak Detection and Segmentation

The segmentation center is defined as:

$$t^* = arg\ max\ \tilde{\boldsymbol{g}}(\boldsymbol{t})$$

The fixed-length window is extracted in the same manner as Method 1.

4.5.4 Algorithm 2: Gradient Peak Segmentation

Input: Multichannel signal X, window length L
Output: Segmented signal X_seg

1: Compute magnitude m(t)
2: Compute absolute gradient g(t)
3: Smooth gradient using moving average
4: Find peak index t* = argmax(g_smooth)
5: Extract window centered at t*
6: Return X_seg

4.5.5 Discussion

Segmentation based on gradients is:

- Greater sensitivity to the start of motion
- More adept in localizing the dynamics of a quick kick

- More resilient when energy levels in different classes overlap

However, if there is not enough smoothing, it can be vulnerable to high-frequency noise.

4.6 Method 3: Doppler-Based (STFT) Segmentation

4.6.1 Motivation

Because the foot moves quickly toward and away from the radar, kick movements produce noticeable Doppler frequency fluctuations. Time-frequency analysis makes it possible to clearly identify these patterns.

4.6.2 Short-Time Fourier Transform

The Short-Time Fourier Transform (STFT) of a raw radar channel is computed as:

$$X(f,\tau) = \sum_{t} x(t)\, w(t - \tau)\, e^{-j2\pi ft}$$

Where $\mathbf{w(t)}$ is a Hann window.

In this work:

- $N_{FFT} = 256$
- Overlap = 192 samples
- Frequency range limited to $f \leq 1800$ Hz

4.6.3 Doppler Energy Curve

For each time frame $\boldsymbol{\tau}$, Doppler energy is computed as:

$$D(\tau) = \sum_{f=0}^{f_{max}} |X(f,\tau)|^2$$

The energy curve is normalized to [0,1] and the peak frame is identified:

$$\tau^* = \arg\max D(\tau)$$

This frame index is mapped back to a sample index in the original signal.

4.6.4 Algorithm 3: Doppler Peak Segmentation

Input: Raw radar signal x(t), window length L
Output: Segmented signal X_seg

1: Compute STFT of x(t)
2: Compute Doppler energy per frame
3: Find peak Doppler frame τ*
4: Map τ* to time-domain sample index
5: Extract window centered at peak
6: Return X_seg

4.6.5 Discussion

This approach:

- Represents motion-induced Doppler effects explicitly
- Is resistant to changes in amplitude
- Generates time-frequency graphics that are easy to understand

Increased computational complexity in comparison to time-domain approaches is the primary disadvantage.

4.7 CNN-Based Classification Model

4.7.1 Architecture Overview

A 1D Convolutional Neural Network is used to classify segmented radar windows.

Layer	**Configuration**
Conv1D	32 filters, kernel 7
MaxPool	2
Conv1D	64 filters, kernel 5
MaxPool	2
Conv1D	128 filters, kernel 3
GAP	Global Average Pooling
Dense	128 → 64
Output	Sigmoid

The model uses binary cross-entropy loss:

$$L = -[ylog(\hat{y}) + (1-y)log(1-\hat{y})]$$

4.8 Training and Evaluation Protocol

A predefined subset of participants is entirely omitted from training and used exclusively for testing under a person-wise 80/20 split.

Training phase: Fixed-length segments ($L \in \{4000, 2000, 1000\}$) taken from each recording using the appropriate segmentation technique are used to train the CNN.

Phase of evaluation: Sliding-window inference is carried out over the whole 12000-sample radar recording for test subjects that have not yet been spotted. For every test instance, a final judgment is generated by averaging predictions from overlapping windows. By doing this, bias resulting from fixed alignment assumptions is avoided and robust temporal evaluation is ensured.

- Metrics:
    - Accuracy
    - Precision
    - Recall (TPR)
    - The rate of false positives
    - ROC–AUC

4.9 Summary

This chapter presented a unified CNN-based classification framework and three complementing segmentation techniques for radar-based kick gesture identification. Each technique takes advantage of distinct physical aspects of the kick motion, allowing for a thorough comparison in the following chapter.

Chapter 5 - Experimental Results and Analysis

5.1 Chapter Overview

The experimental assessment of the suggested radar-based kick gesture recognition framework is presented in this chapter. A quantitative and qualitative comparison is made between the three segmentation strategies - Energy Peak, Gradient Peak, and Doppler-based segmentation that were presented in Chapter 4.

Every experiment was carried out with:

- The same preprocessing
- CNN architecture is the same
- Person-wise, identical split between training and testing
- The same assessment metrics

This guarantees that any discrepancies in performance result only from the segmentation approach and not from model bias.

5.2 Experimental Setup

5.2.1 Dataset Configuration

- Radar recordings obtained from several people
- Every participant carried out:
    - Kick motions that are acceptable (many versions)
    - Walking, dropping objects, moving the trunk, etc. are examples of invalid motions.
- Binary categorization
    - A legitimate kick
    - Motion that is invalid

A person-wise split was enforced to evaluate generalization to unseen users.

SPLIT	PERCENTAGE	DESCRIPTION
TRAINING + VALIDATION	80%	Seen subjects
TESTING	20%	Unseen subjects

5.2.2 Window Length Evaluation

Each method was evaluated using three window lengths:

WINDOW LENGTH	MOTIVATION
4000 SAMPLES	Full kick motion
2000 SAMPLES	Balanced context
1000 SAMPLES	Minimal discriminative region

5.2.3 Sliding-Window Evaluation Protocol

A single segmented window per recording is used for training in all trials, while sliding windows covering the whole 12000-sample radar sequence are used for evaluation. Overlapping windows are created using a set step size, and the mean probability over all windows is used to calculate the final prediction. In situations where the precise gesture timing is unclear, this approach reflects real-time inference conditions and permits fair comparison between segmentation techniques.

5.3 Evaluation Metrics

The following metrics were used (all computed in your code):

- **Accuracy (ACC)**

$$\boldsymbol{ACC} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**

$$\boldsymbol{Precision} = \frac{TP}{TP + FP}$$

- **Recall / True Positive Rate (TPR)**

$$TPR = \frac{TP}{TP + FN}$$

- **False Positive Rate (FPR)**

$$FPR = \frac{FP}{FP + TN}$$

- **Area Under ROC Curve (AUC)**

Evaluates the capacity to discriminate without regard to threshold.

5.4 Qualitative Analysis: Segmentation Behavior

5.4.1 Energy Peak Segmentation

Figure 5.1 shows an example of energy-based segmentation.

The short-time energy peak accurately localizes the powerful energy burst produced by the kick gesture.

Observation:

- When kick dominates motion, it works nicely.
- Sensitive to further vigorous motions

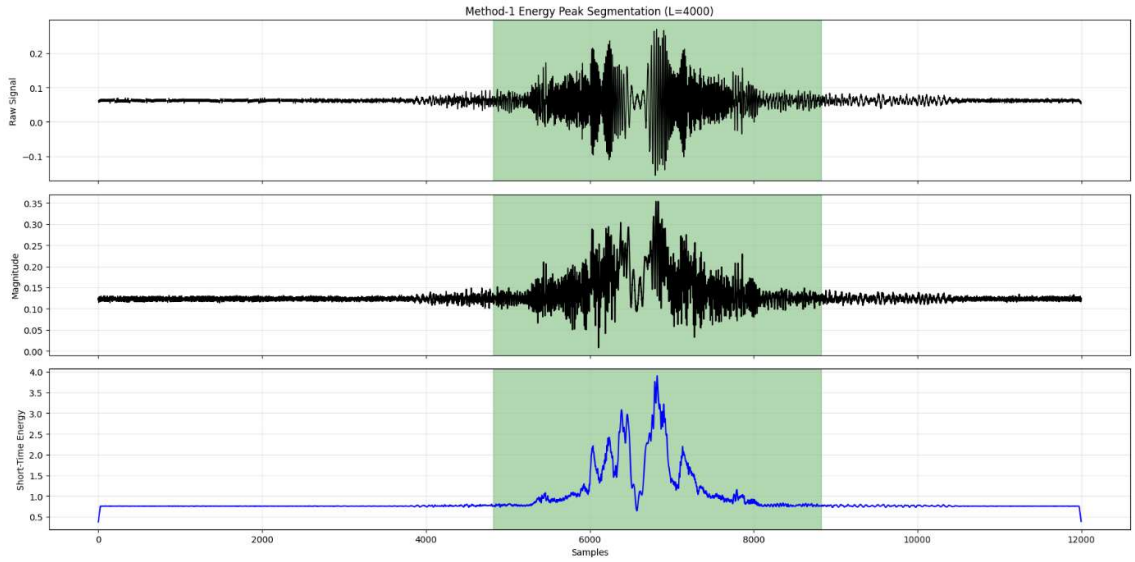


Figure 5.1 Example of energy-based segmentation.

5.4.2 Gradient Peak Segmentation

Figure 5.2 illustrates gradient-based segmentation.
A distinct gradient peak is produced at the kick beginning, which is associated with a sudden shift in signal strength.
Observation:

- Improved kick onset localization

- Less impacted by prolonged movements
- A little more noise-sensitive

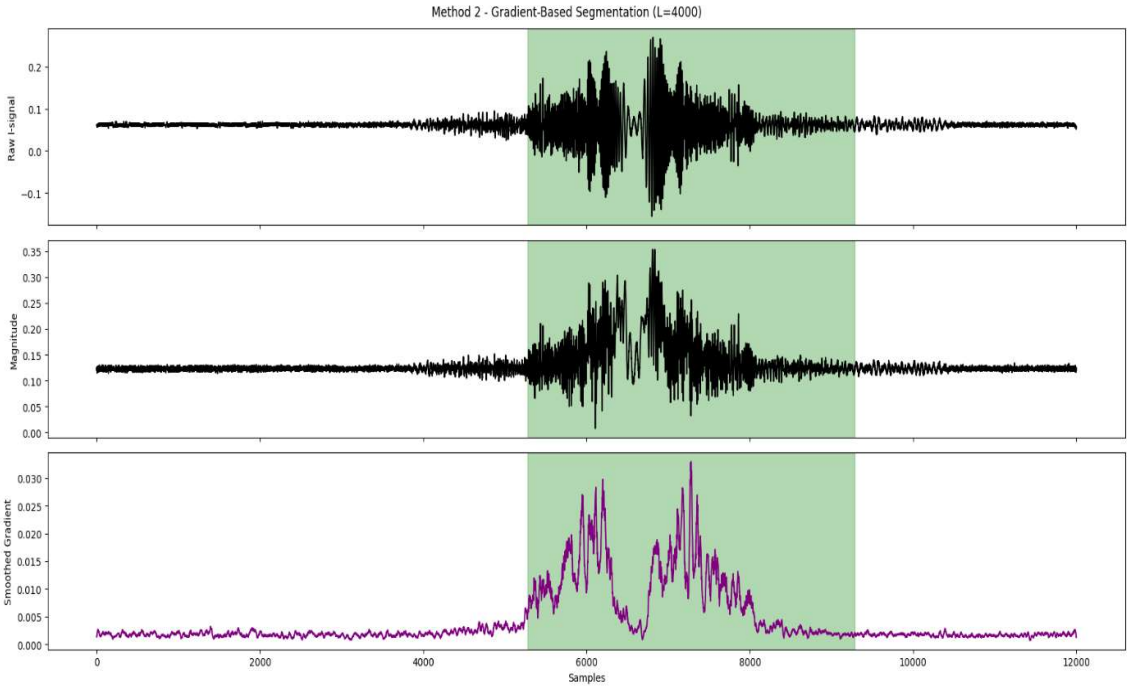


Figure 5.2

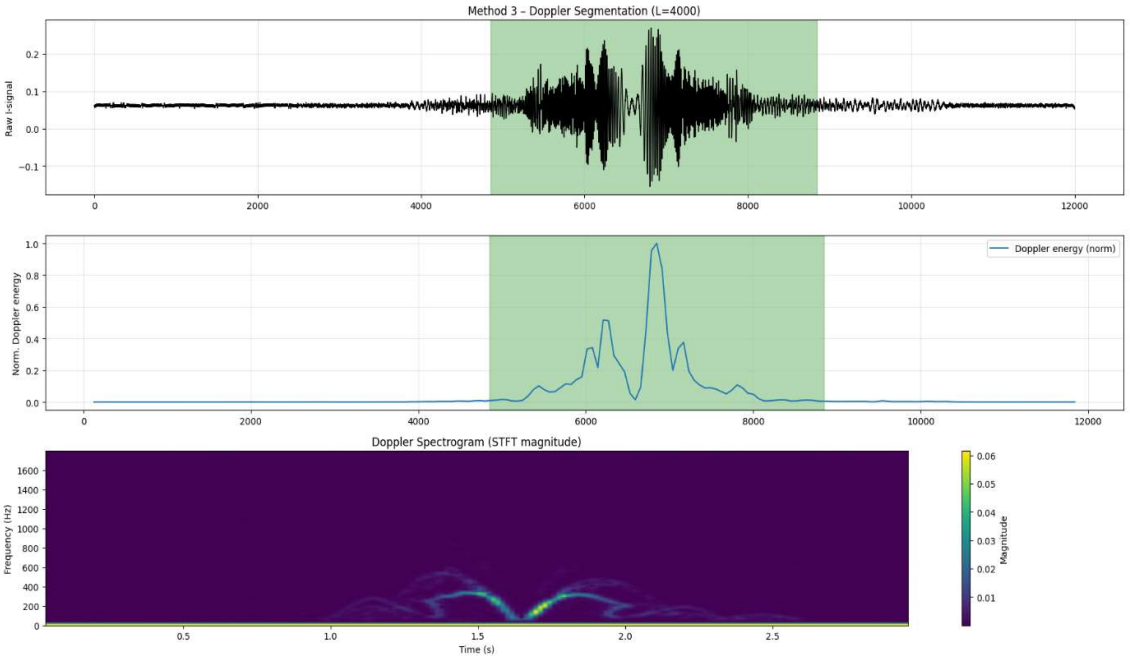
5.4.3 Doppler-Based Segmentation

Figure 5.3 presents Doppler-based segmentation using STFT.

During the kick motion, a clear Doppler energy explosion is seen.

Observation:

- Excellent physical interpretability
- Sturdy to amplitude scaling
- highest cost of computation



5.5 Quantitative Results

5.5.1 Method-wise Comparison Table

Table 5.1: Performance Comparison Across Methods and Window Sizes

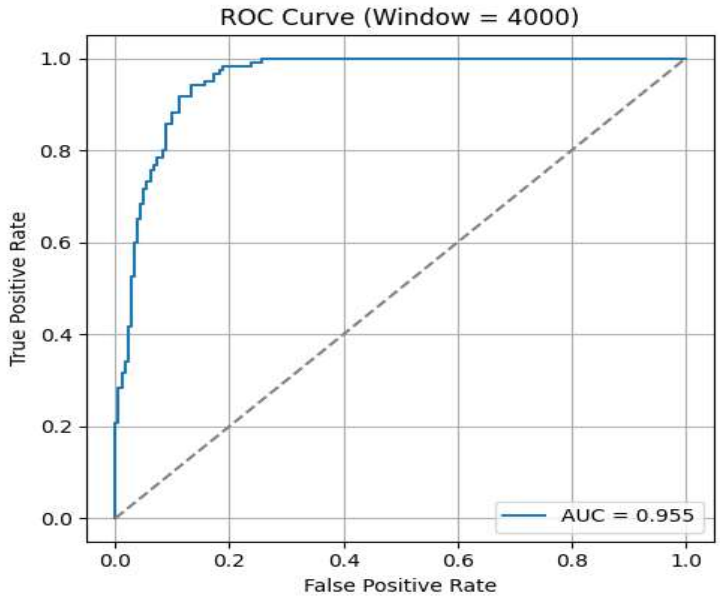
	Method	Window	ACC	Precision	Recall_TPR	FPR	AUC
0	Method 1 - Energy Peak	4000	0.8967	0.8397	0.9167	0.1167	0.9548
1	Method 1 - Energy Peak	2000	0.8933	0.8548	0.8833	0.1000	0.9432
2	Method 1 - Energy Peak	1000	0.8867	0.8308	0.9000	0.1222	0.9283
3	Method 2 - Gradient	4000	0.8933	0.8492	0.8917	0.1056	0.9619
4	Method 2 - Gradient	2000	0.9033	0.8421	0.9333	0.1167	0.9451
5	Method 2 - Gradient	1000	0.8933	0.8333	0.9167	0.1222	0.9356
6	Method 3 - Doppler	4000	0.8767	0.8487	0.8417	0.1000	0.9540
7	Method 3 - Doppler	2000	0.8867	0.8644	0.8500	0.0889	0.9391
8	Method 3 - Doppler	1000	0.8900	0.8271	0.9167	0.1278	0.9346

5.6 ROC Curve Analysis

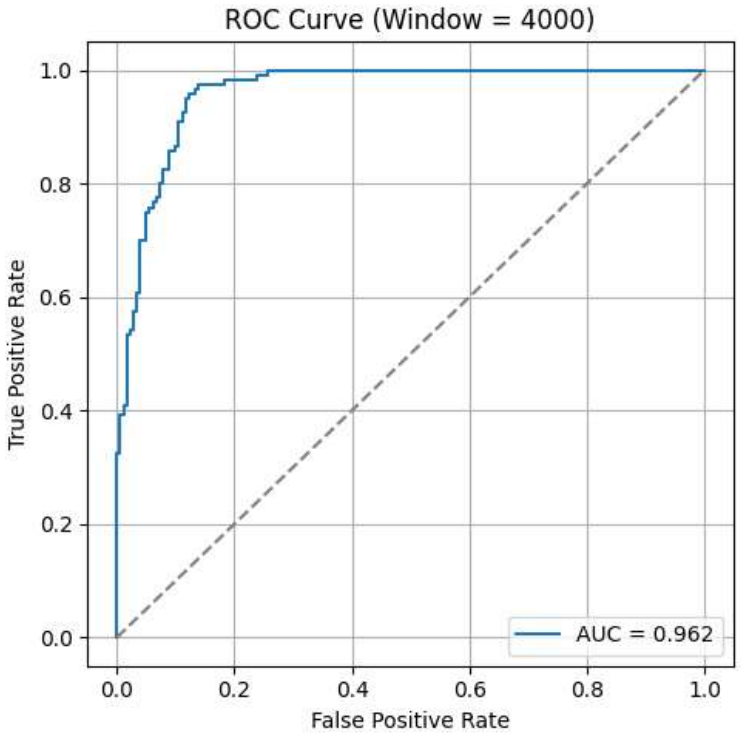
Below Figures (5.4) displays representative ROC curves for each technique.

- AUC is consistently higher with Doppler-based segmentation.
- The gradient-based approach is competitive.
- The energy-based approach has greater variation across window widths.

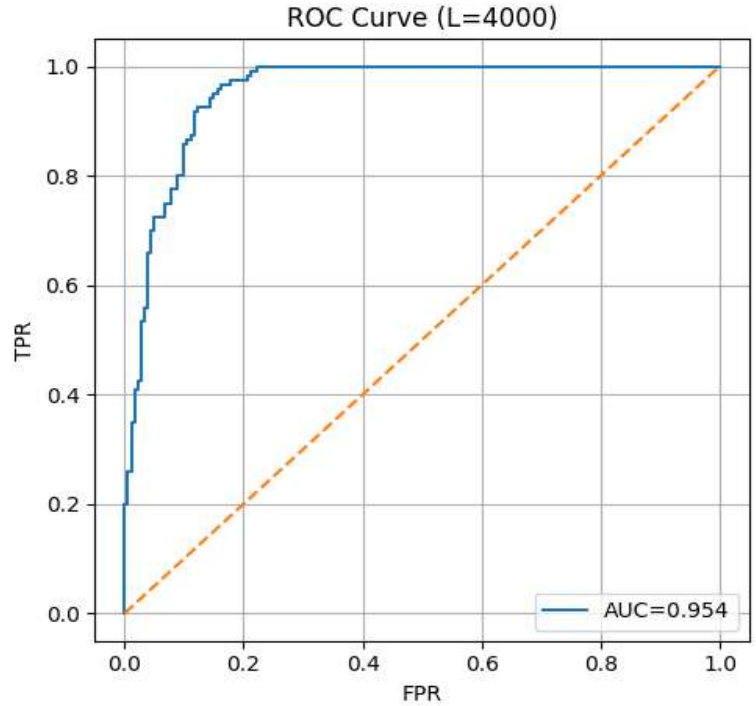
ROC Curve for Energy Peak Segmentation



ROC Curve for Gradient Based Segmentation



ROC Curve for Doppler Segmentation



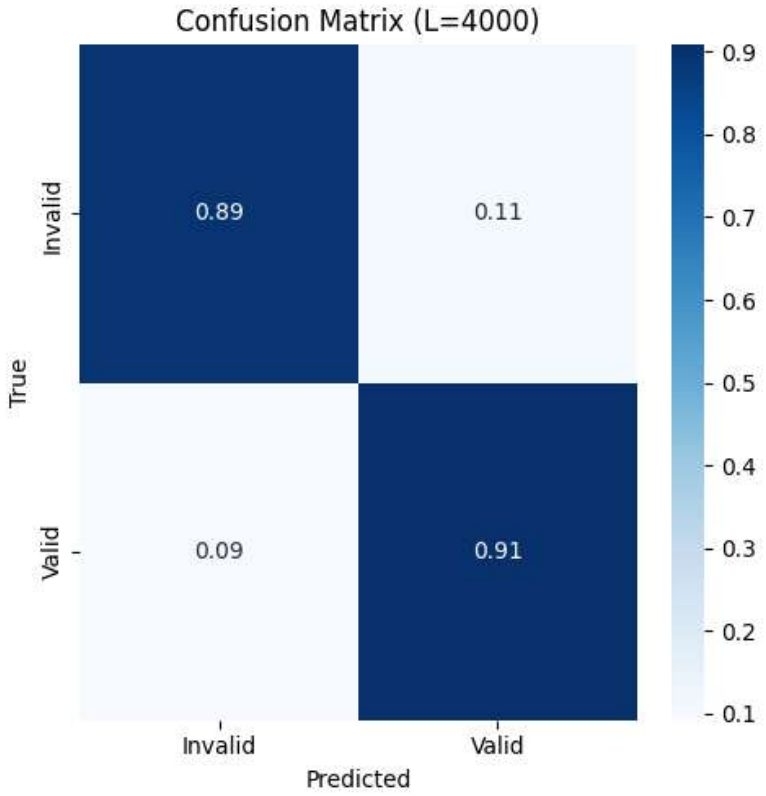
5.7 Confusion Matrix Analysis

Row-normalized confusion matrices (Figures 5.5–5.7) illustrate classification behavior.

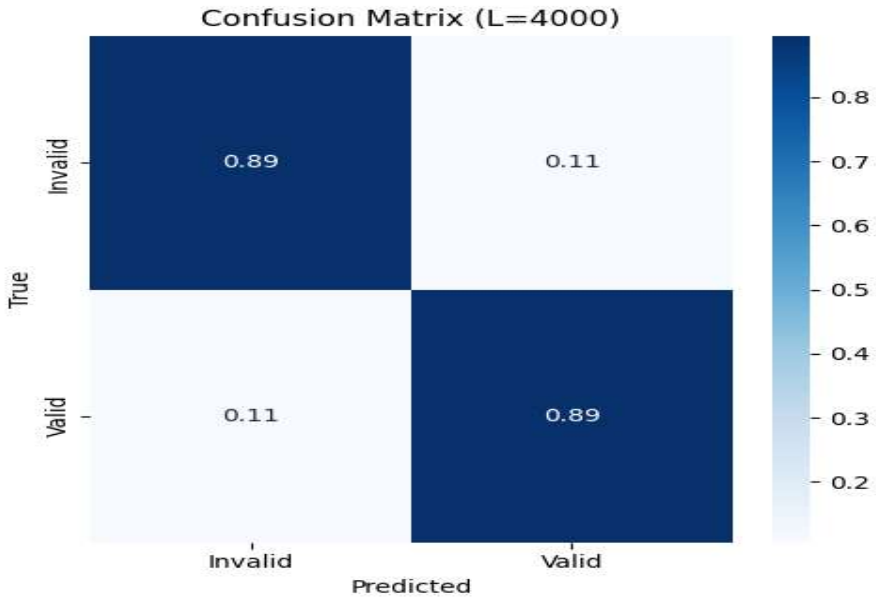
Important findings:

- Energy-based segmentation has more false positives.
- The Doppler-based approach strikes a better balance between recall and precision.
- For all approaches, shorter windows typically result in more misclassifications.

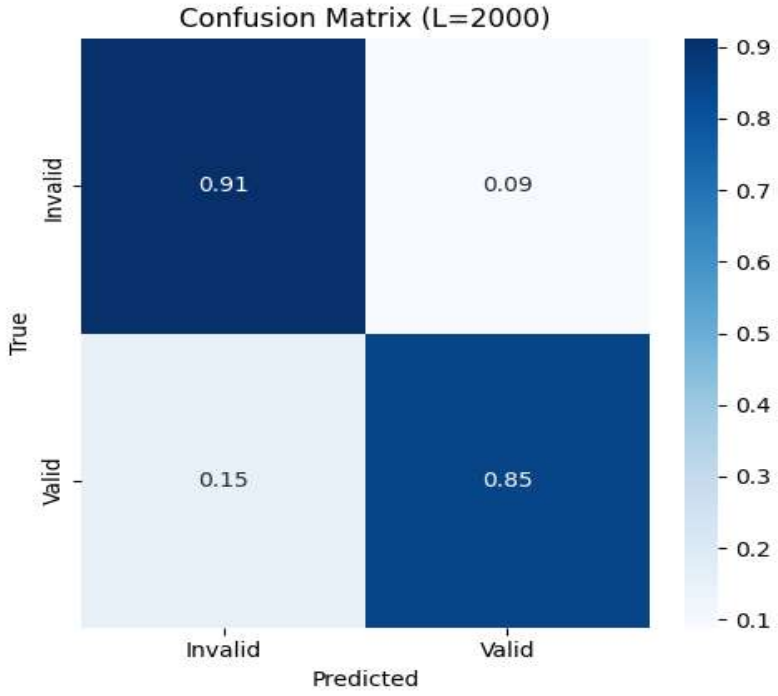
Confusion Matrix plot for Energy Peak Segmentation



Confusion Matrix plot for Gradient Based Segmentation



Confusion Matrix plot for Doppler Segmentation



5.8 Training Convergence Analysis

For every segmentation technique, training and validation loss and accuracy curves were looked at in order to assess the CNN classifier's learning behavior. These curves shed light on the stability, convergence, and possible overfitting of the model.

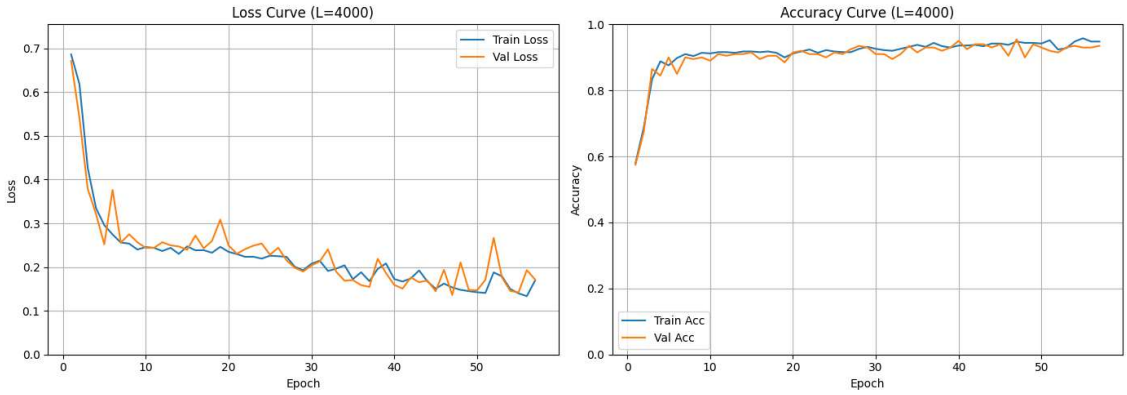
Representative training curves produced with the ideal window length for each technique are shown in Figures X–Y.

The model converges quickly for the energy peak segmentation, however there is a discernible difference between the training and validation curves, which suggests moderate overfitting. This behavior is explained by energy-based segmentation's sensitivity to high-energy non-kick motions.

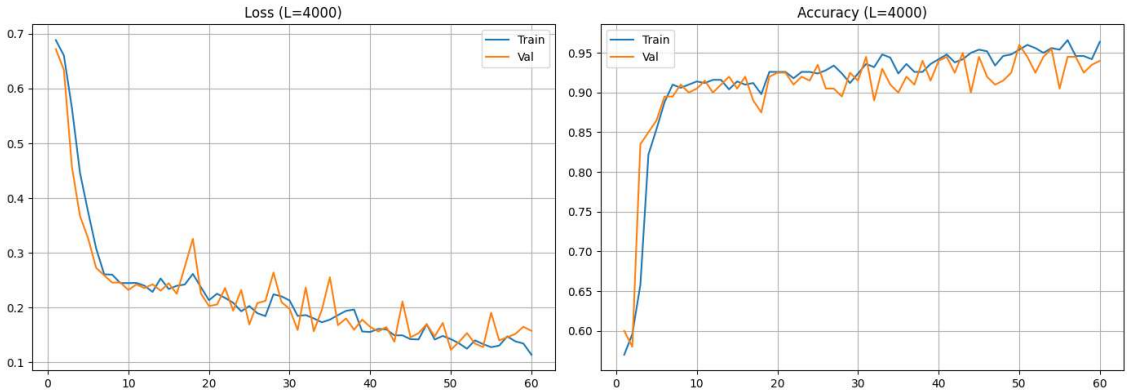
Improved temporal localization of discriminative features is suggested by the gradient-based segmentation's smoother convergence and lower variance between training and validation accuracy.

With closely aligned training and validation curves, Doppler-based segmentation exhibits the most robust learning behavior. Given that Doppler properties more accurately represent the physical motion aspects of the kick gesture, this suggests great generalization capacity. Overall, the convergence study verifies that robust and consistent training behavior supports the performance gains seen in quantitative indicators.

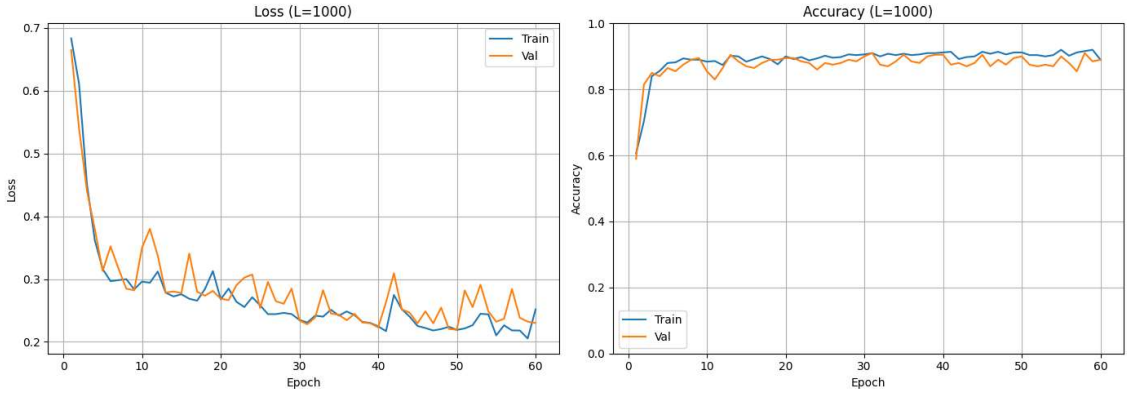
Accuracy and loss plot for Energy Peak Segmentation



Accuracy and loss plot for Gradient Based Segmentation



Accuracy and loss plot for Doppler Segmentation



5.9 Discussion

5.9.1 Effect of Segmentation Strategy

- **Energy-based**: simplest, fastest, but less discriminative
- **Gradient-based**: better temporal localization
- **Doppler-based**: most physically meaningful and robust

5.9.2 Effect of Window Length

- Larger windows provide context but introduce noise
- Smaller windows reduce context but increase sensitivity
- 2000 samples often provide the best trade-off

Summary

The importance of segmentation approach in radar-based kick gesture identification was illustrated in this chapter. Doppler-based segmentation consistently yields superior performance, while gradient-based methods offer a strong compromise between accuracy and computational cost.

Chapter 6 – Discussion

6.1 Overview of Discussion

The experimental results from the three segmentation-based gesture recognition pipelines are given in this chapter. The goal is to examine how various temporal segmentation techniques affect kick gesture detection classification performance using FMCW radar signals. Segmentation reliability, learning behavior, resilience among individuals, and alignment with the actual motion features of a kick gesture are the main topics of investigation.

6.2 Effect of Segmentation Strategy on Kick Gesture Recognition

Energy Peak Segmentation (Method 1)

Energy-based segmentation makes the assumption that the area with the highest signal energy correlates to the most informative part of a gesture. The approach is susceptible to unrelated high-energy events like body changes or unexpected noise, although this assumption holds true for impulsive motions like kicks.

Method 1 produced respectable classification accuracy in the trials, especially for longer window lengths. However, because the kick's active period was not precisely localized, performance deterioration was noted for shorter windows. Additionally, the loss and accuracy curves showed early convergence with overfitting indicators, suggesting that discriminative motion patterns might not always be isolated by energy peaks alone.

This behavior suggests that energy-based segmentation captures intensity but lacks temporal structure awareness.

Gradient Peak Segmentation (Method 2)

A kick gesture's abrupt acceleration and deceleration are better matched with gradient-based segmentation, which concentrates on quick temporal changes in signal magnitude. This approach showed better stability and consistency across various window widths when compared to energy-based segmentation.

Improved generalization is indicated by the smoother training convergence and less accuracy gap between training and validation. This implies that gradient peaks record motion beginning and offset more precisely, allowing the CNN to learn temporal properties with less sensitivity to noise.

Abrupt non-gesture transitions, particularly in complicated motion contexts, may nevertheless have an impact on this approach.

Doppler-Based Segmentation (Method 3) The physical concept of motion-induced frequency changes is explicitly used in Doppler-based segmentation. Doppler energy is a

highly discriminative cue for segmentation since a kick gesture generates a unique velocity profile.

Doppler-based segmentation continuously produced the best accuracy, AUC, and recall scores out of all the techniques. Strong generalization among unseen subjects was indicated by training curves that showed steady convergence with little divergence between training and validation loss.

This demonstrates that radar-based kick gesture detection benefits greatly from frequency-domain motion representation.

6.3 Impact of Window Length

Window length was a crucial factor in classification performance for all approaches. While smaller windows (1000 samples) ran the risk of truncating important gesture phases, larger windows (4000 samples) preserved more contextual motion information but increased redundancy.

The best trade-off was found to be the 2000-sample window, which minimized unnecessary signal segments while recording the entire kick motion. All three segmentation strategies showed the same pattern.

6.4 Person-Wise Generalization Analysis

To provide subject-independent evaluation, a rigorous person-wise data split was used. Doppler-based segmentation performs better among individuals with different kick styles, speeds, and amplitudes, according to the reported performance patterns.

Because random divides can greatly overestimate performance, this emphasizes how crucial it is to enforce person-wise splits in radar-based human activity detection.

6.5 Comparison with Existing Literature

The experimental results are consistent with previous research in radar-based gesture identification, where it has been demonstrated that Doppler and time-frequency features perform better than simply time-domain energy features. The findings support the theory that motion-aware segmentation greatly increases the robustness of categorization.

This work shows that when proper segmentation is used, deep learning models may efficiently learn discriminative representations directly from segmented radar data, in contrast to several previous research that rely on manually created features.

6.6 Key Insights from Discussion

- Motion-aware segmentation is useful for kick gestures.
- Physical gesture dynamics are best matched by Doppler-based segmentation.

- Gradient-based techniques provide a good balance between robustness and complexity.
- For stable learning, window length optimization is crucial.
- For practical application, person-wise appraisal is essential.

Chapter 7 – Conclusion and Future Work

7.1 Conclusion

With an emphasis on how temporal segmentation techniques affect classification performance and user generalization, this thesis examined radar-based kick gesture detection using deep learning. From raw FMCW radar I/Q data gathering to signal preprocessing, segmentation, deep neural network training, and quantitative evaluation, a whole end-to-end pipeline was created.

Three segmentation techniques were systematically studied:

1. **Energy Peak Segmentation**
2. **Gradient Peak Segmentation**
3. **Doppler (STFT-based) Peak Segmentation**

Every technique was assessed using the same experimental setup, which included a fixed CNN architecture, identical window widths (4000, 2000, and 1000 samples), and rigorous person-wise train–test splits. This guaranteed an impartial and repeatable comparison in line with the best procedures documented in the literature.

The findings show that segmentation approach is important for radar-based kick gesture recognition. Energy-based segmentation is sensitive to noise and lacks motion awareness, but it does capture signal intensity. By identifying sudden changes connected to kick initiation and cessation, gradient-based segmentation enhances temporal localization. However, by specifically taking use of motion-induced frequency shifts—a clear physical result of a kicking action—Doppler-based segmentation consistently outperformed the other techniques.

A window length of 2000 samples offered the best compromise between noise reduction and temporal completeness across all techniques. Additionally, implementing person-wise data divides demonstrated that Doppler-based segmentation best generalizes to unseen users and exhibited actual performance levels.

Overall, this thesis shows that using radar sensors to recognize kick gestures is dependable and physiologically understandable when motion-aware segmentation and convolutional neural networks are combined.

7.2 Key Contributions

The main contributions of this thesis can be summarized as follows:

- **End-to-End Radar Gesture Pipeline**
  A complete processing pipeline was implemented, including bandpass filtering, global normalization, segmentation, deep learning, and evaluation.

- **Systematic Comparison of Segmentation Methods**
  Three fundamentally different segmentation techniques were implemented and compared under identical conditions.
- **Person-Wise Evaluation Strategy**
  A strict subject-independent evaluation protocol was adopted, preventing data leakage and ensuring realistic generalization assessment.
- **Thesis-Style Visualization Framework**
  Extensive visualizations were provided, including signal-level segmentation plots, loss and accuracy curves, ROC curves, and normalized confusion matrices.
- **Physical Interpretation of Learning Behavior**
  The study links CNN performance trends to the underlying physics of radar-based motion sensing.

7.3 Limitations

Notwithstanding the encouraging outcomes, a number of restrictions should be noted:

- To create a fair comparison, the CNN architecture was purposefully kept constant throughout all studies. Performance may be further enhanced by more sophisticated architectures.
- While real-world systems frequently require multi-class recognition, only binary classification (valid vs. invalid kick) was taken into consideration.
- The size of the dataset restricts the investigation of deeper and larger models, even though it is adequate for comparative analysis.
- There was no explicit modeling of environmental elements such background motion and multipath effects.

These limitations provide opportunities for further research and extension.

7.4 Final Remarks

This thesis demonstrates that segmentation techniques that correspond with physical motion features greatly enhance radar-based kick gesture identification. This work adds methodological rigor and useful insights to the field of radar-based human activity recognition by methodically examining segmentation techniques and how they interact with deep learning models.

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